

Program 8 : Create Kubernetes Pods using YAML Descriptors

Project Structure

```
.
├── nodejs-configmap.yml
├── nodejs-pod.yml
└── server.js
```

Step 1 : Start the minikube

minikube start—driver=docker

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ minikube start --driver=docker
🐸 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌞 Enabled addons: storage-provisioner, default-storageclass, dashboard

! /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
   ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏁 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$
```

Step 2: Create a node js app

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ cat server.js
const http= require("http");
const port = 3001;
const server = http.createServer((req,res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 3 : create a nodejs-configmap.yaml file

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http= require("http");
    const port = 3001;
    const server = http.createServer((req,res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 4 : create a nodejs-pods.yaml file.

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ cat nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
  - name: nodejs
    image: node:18
    command: ["node", "/usr/src/app/server.js"]
    volumeMounts:
    - name: app-code
      mountPath: /usr/src/app
    ports:
    - containerPort: 3001
  volumes:
  - name: app-code
    configMap:
      name: nodejs-app-config
```

Step 5 : Apply the yaml files

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ k apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
```

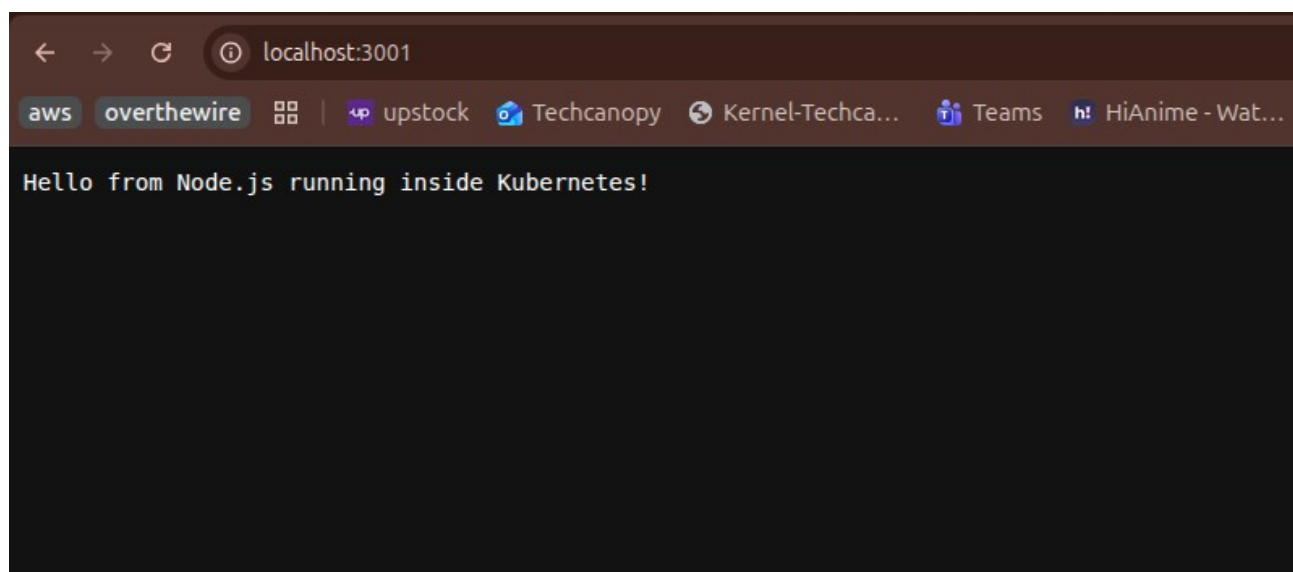
```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ k apply -f nodejs-pod.yaml
pod/program-8 created
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$
```

Step 6 : Forward the port and access the nodejs app

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ k get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-c78kt  1/1     Running   1 (8m23s ago)  19m
nginx-deployment-6f9664446b-vq5k8  1/1     Running   1 (8m23s ago)  19m
program-8                           1/1     Running   0           2m5s
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$

1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab8$ k port-forward pod/program-8 3001:3001
Forwarding from 127.0.0.1:3001 -> 3001
Forwarding from [::1]:3001 -> 3001
Handling connection for 3001
Handling connection for 3001
```

Step 7 : Check the out at <http://localhost:3000>



Step 8 : Delete Resources

```
kubectl delete pods --all
kubectl delete svc --all
kubectl delete deployment --all
```